A Two-Dimensional Smoothed Particle Hydrodynamics Solver

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October 28, 2026

Abstract

This paper documents the design and evaluation of a small two-dimensional smoothed particle hydrodynamics (SPH) solver. The implementation combines particle-based fluid dynamics, radius-based neighbor search, numerical integration, and an interactive renderer.

Contents

1 Introduction **1**

2 Research Questions **1**

3 Methodology **2**
3.1 SPH Approximation 2
3.2 Software Structure 2
3.3 Experimental Setup 2

4 Results **2**
4.1 Correctness Tests 2
4.2 Discussion 2

5 Conclusion **3**

1 Introduction

Particle-based methods are useful for simulating fluids with free surfaces and changing topology. Smoothed particle hydrodynamics represents a continuum by a set of moving particles and approximates field quantities using a smoothing kernel [1, 2].

The goal of this project is to implement and inspect a compact two-dimensional SPH solver in C++17. The application includes density and pressure estimation, viscosity, gravity, boundary particles, and an interactive SFML renderer. This paper describes the numerical model, the software architecture, and the tests used to assess the implementation.

2 Research Questions

The evaluation focuses on three questions:

1. Does the solver preserve the expected qualitative behavior of a small fluid scene?

2. How does the radius-based neighbor search affect the simulation's computational work?
3. Which implementation parameters have the greatest visible effect on stability and motion?

3 Methodology

3.1 SPH Approximation

For a quantity A evaluated at particle i, the SPH approximation is written as

$$A_i \approx \sum_j m_j \frac{A_j}{\rho_j} W(\|\mathbf{r}_i - \mathbf{r}_j\|, h), \tag{1}$$

where m_j is the particle mass, ρ_j is the density, h is the smoothing radius, and W is a compact-support kernel.

The implementation first constructs neighbor lists for particles within the smoothing radius. It then estimates density, computes pressure from the density error relative to the rest density, and combines pressure, viscosity, and gravity forces before integrating position and velocity.

3.2 Software Structure

The solver is separated into the following components:

- `Particle` stores particle state and physical properties.
- `NeighborSearch` constructs local particle neighborhoods.
- `SPHSolver` performs density, force, and integration steps.
- `Renderer` displays the scene and exposes runtime controls.

This separation allows the numerical code and neighbor-search behavior to be tested without requiring an interactive graphics window.

3.3 Experimental Setup

Experiments use the startup scene defined in `src/main.cpp`. The scene contains movable particles and fixed particles forming a simple container. The main parameters are the rest density, gas constant, viscosity, smoothing radius, timestep, and gravitational acceleration.

4 Results

4.1 Correctness Tests

The repository includes tests for neighbor search and solver behavior. The test suite is executed with CTest and provides a reproducible check for core invariants such as neighborhood membership and particle-state updates.

4.2 Discussion

The final evaluation should report measurements for several parameter settings and include representative screenshots or plots. Figures can be added to this section with the `graphicx` package once experiment output is available.

Table 1: Planned evaluation metrics.

Metric	Purpose
Neighbor count	Assess local interaction construction
Density deviation	Measure incompressibility behavior
Maximum velocity	Detect unstable parameter settings
Frame time	Estimate interactive performance

5 Conclusion

This project provides a small and inspectable SPH fluid solver with an interactive visualization and isolated tests for its core components. The separation between neighbor search, numerical integration, and rendering gives the implementation a useful basis for experiments.

Future work could compare different smoothing kernels, improve boundary handling, and benchmark spatial acceleration structures with larger particle sets.

References

[1] Leon B. Lucy. A numerical approach to the testing of the fission hypothesis. *The Astronomical Journal*, 82:1013–1024, 1977.

[2] Joseph P. Morris, Patrick J. Fox, and Yi Zhu. Modeling low reynolds number incompressible flows using sph. *Journal of Computational Physics*, 136(1):214–226, 1997.